A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Solid wastes - Set 1

Mark scheme

AQA spec 3.4.3.2.12; Pollution; textbook pp. 279-285

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Landfill fell from 79% to 14% (large decrease) **1** Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Organic matter decays anaerobically releasing methane (and CO2) - greenhouse gases **1** Toxic leachate may leak into groundwater / rivers from poorly managed sites **1** Uses large areas of land – loss of habitats / farmland – contamination prevents later use **1** Resource value of metals, glass, plastics is lost **1** Transport noise / congestion / litter / pests / odour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
01.3	**Any 4 of 7** 1 mark each for any 4 of the 7 points below, in any order, maximum 4. **1** Polymer / clay liner to prevent escape of leachate **1** Collection and treatment of leachate **1** Collection of methane and use as a fuel **1** Impermeable cap (clay and polymer) when complete to stop water entering **1** Regular covering with soil (pests, odour) – deodorising sprays **1** Perimeter fence to trap wind-blown litter **1** Separation and recording of waste types – reactive / flammable wastes not dumped together **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste • **Impermeable** → does not let water through **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	4
01.4	1 mark for the point below. **1** Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	1

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	
02.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Volume of ash is much less than the original waste (less landfill) **1** Heat used for district heating / electricity generation (energy recovery) **1** No sorting / complicated management needed **1** Destroys pathogens / flammable organic wastes converted to CO2 and water **1** Refuse-derived fuel can replace fossil fuels (eg cement kilns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
02.2	**1 mark each** 1 mark for each of the 4 points below. **1** High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) **1** Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone **OR** removes heavy metals for storage at high pH **1** Wet waste needs extra fuel to burn and lowers combustion temperature – composting recovers its nutrient value **1** Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Ash may contain concentrated heavy metals / toxic residues **1** Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

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Question	Marking guidance	Marks
02.4	**Any 1 of 2** 1 mark each for any 1 of the 2 points below, in any order, maximum 1. **1** Fuel (natural gas) needed to burn wet / non-flammable wastes **1** Cost of treating effluent gases (SO2, NOx, HCl, smoke) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Asbestos:** fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma – disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) **2 Cyanide:** very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested – disposal: incineration so carbon and nitrogen are separated and oxidised to CO2 and NOx (much smaller problem) **2 High-level radioactive waste:** intense ionising radiation, heat generation, long half-lives – disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **vitrification** → sealing waste into glass **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
03.2	1 mark for the point below. **1** The fibres are bound / cannot become airborne so cannot be inhaled **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material **1** The solid cement immobilises the waste so it cannot leak / leach out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

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Question	Marking guidance	Marks
03.4	1 mark for the point below. **1** Even if the glass shatters the waste stays encapsulated in the fragments – glass is more stable / resistant to heat and leaching over very long timescales **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **vitrification** → sealing waste into glass • **leaching** → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
04.1	**1 mark each** 1 mark for each of the 6 points below. **1** Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes **1** Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH **1** Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow **1** Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser **1** Unattractive topography - landscaping / reshaping **1** Dust / heavy metals - cover and keep at high pH **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** So that plant roots cannot reach the toxic layer **1** Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions **1** Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them **1 Built-in obsolescence:** products designed to have short lives / wear out / go out of fashion so customers upgrade **1** Disposable products (razors, lighters, pens) that must be replaced **1** Over-packaging to make products attractive **OR** increases wastes from manufacture and mineral extraction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.2	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1 Properties of the waste:** toxicity, flammability, degradability, radioactivity **1** Population density (collection costs) **1** Land availability for landfill **1** Availability of recycling technology / markets **1** Legislation / regulatory framework (landfill tax) **1** Public willingness to recycle / environmental awareness **1** Processing costs / household income **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** aerobic decomposition produces little methane – nutrients / organic matter returned to soil (circular economy) – reduces landfill volume **1 Disadvantage:** needs separation / collection of clean organic waste – odour / pests / pathogens if poorly managed – low-value product – not suitable for mixed waste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2

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Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
06.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

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